

Submission Statement

Liulab 1st Scientific Figure Design Competition

Background and Design Goal

This submission redesigns Figure 2 to improve visual hierarchy, readability, and narrative coherence while keeping all underlying data and statistical conclusions unchanged.

The original Morandi-style figure was already stable and elegant, but had limited visual impact. Therefore, I prepared two parallel variants:

- **Version A (Morandi, refined):** conservative optimization with clearer structure.
- **Version B (High-contrast, clash palette):** stronger visual impact for emphasis and communication.

Main Revisions (Version A: Morandi, Refined)

- **Panels A/B:** The overall structure was largely preserved. Panel B transparency was slightly tuned to improve curve legibility and reduce visual masking.
- **Panels C/D:** Panel sizing and visual language were unified. The two distributions were moved closer for a tighter composition, side whitespace was balanced, mean-connecting lines were removed, and p-value annotation style was standardized.
- **Panel E:** The size ratio between the main plot and insets was rebalanced, legend placement was improved, and the lower-right inset was redesigned for clearer group-level interpretation with explicit weak-tail (weak-connection) emphasis.
- **Panel F:** The panel was redesigned as a raincloud-style plot consistent with Panels C/D/E, with explicit significance annotations for direct statistical readability.

Version B (High-Contrast) and Motivation

Version B was generated as a non-overwriting alternative to Version A. The motivation is a deliberate contrast in design language: Morandi palettes are robust and low-risk, while clash palettes require stronger aesthetic control but deliver higher visual salience for key comparisons.

Color Card (Version B)

Category	Hex	Usage
Divergent	#00B3FF	condition color
Convergent	#FF006E	condition color
Random	#FFBE0B	condition color
Coherent	#3A86FF	condition color
Weak-edge FC	#8338EC	decoder panel accent
Shuffle	#FFBE0B	decoder baseline condition

Summary

This work provides a dual-track figure strategy: a publication-safe refined style (Version A) and a high-impact exploratory style (Version B), enabling direct comparison between conservative and expressive scientific design choices.